AGRIVIVE / MEDH-LOCK

UTTAR PRADESH / BUNDING-INSPIRED / INTERLOCKING FIELD EDGE

SERIES 26 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



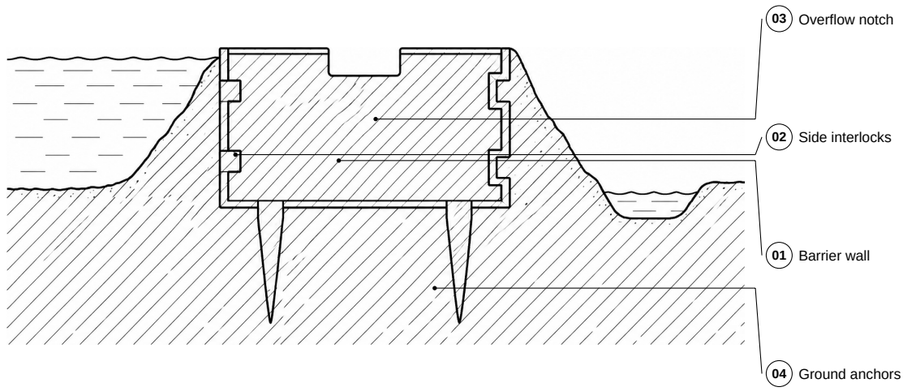
STATE CONTEXT / INDIGENOUS INSPIRATION

Field bunding is studied for soil and water conservation in Uttar Pradesh's Bundelkhand region. MEDH-LOCK proposes a short interlocking barrier with a defined spill notch; soil strength and discharge require site trials.

Practice source: ICAR-CAFRI: Parasai-Sindh watershed, Jhansi | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 160 span x 100 barrier H x 30 thick

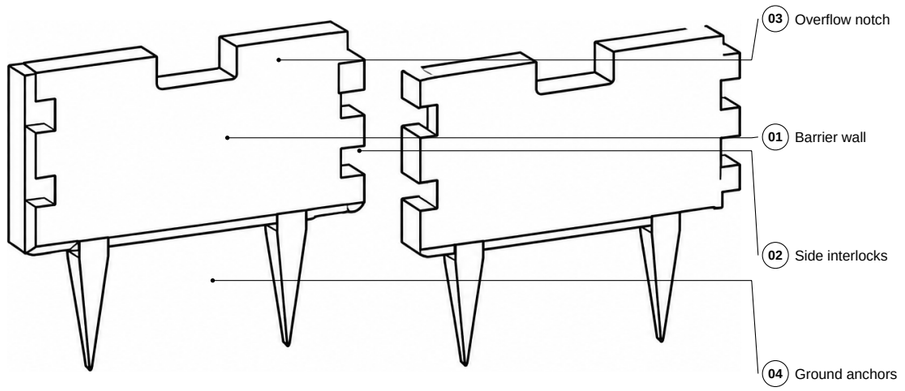


DESIGN / OPERATION NOTES

Align modules in a shallow field bund groove.
Seat the anchors and engage adjacent side joints.
Keep the notch open; overflow passes above its sill, outside the solid wall.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
160 span x 100 barrier H x 30 thick

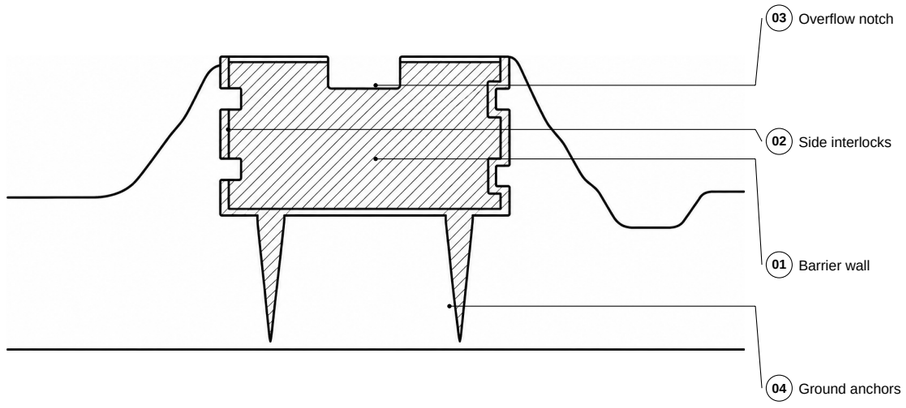


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

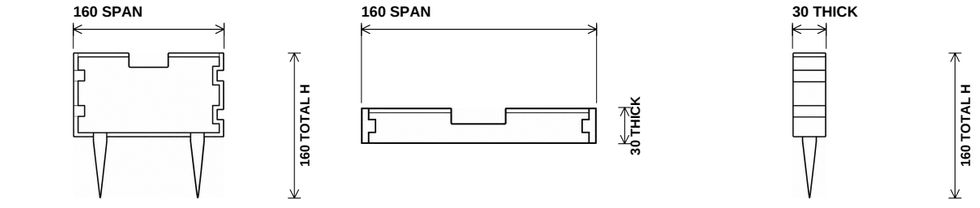
SECTION BASIS / mm
60 anchor extension; overflow notch40 W x20 H



DESIGN / OPERATION NOTES

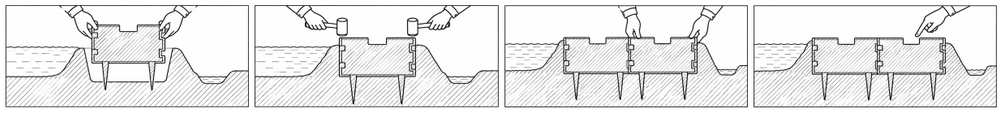
Seat the anchors and engage adjacent side joints.
Keep the notch open; overflow passes above its sill, outside the solid wall.

03 ORTHOGRAPHIC STUDIES / mm



Module 160 span x 100 barrier H x 30 thick; anchors 60 additional. Notch 40 W x 20 deep; total H 160.

05 INSTALLATION / OPERATING SEQUENCE



1 ALIGN MODULE **2 SET ANCHORS** **3 ENGAGE JOINT** **4 CHECK NOTCH**

Prototype fit, access and operation require trials.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Module span / barrier H	**160 / 100**
Wall / anchor extension	**30 / 60**
Notch W x H	**40 x 20**

02 GEOMETRIC CHECK

OVERFLOW NOTCH SECTION

800.0 mm2

40 x 20
No overflow discharge or soil-strength claim

03 PARTS / QTY

01 Barrier wall	**1**
02 Side interlocks	**Integral**
03 Overflow notch	**Integral**
04 Ground anchors	**2 integral**

Wall/anchors/interlock: candidate polymer. Print orientation and wet-soil loads require trials.

04 INTERFACES / CHECKS

Test anchor pull-out and runoff scour.

Check interlock clearance and bypass safety.

Establish overflow behaviour by site measurements.

05 DATUMS / DRAWING BASIS

A: wall base. B: panel centre plane. Anchors extend 60 below A.

PRELIMINARY / VALIDATION PENDING

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-26 / P01	MEDH-LOCK	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	26 / 28